

Components, Sensors, and Circuit Access

Now that we have a little more foundation, let's return to our circuit, and dive a little deeper into the circuit components that you may come across and how you may interact with them (Fig. 9.1).

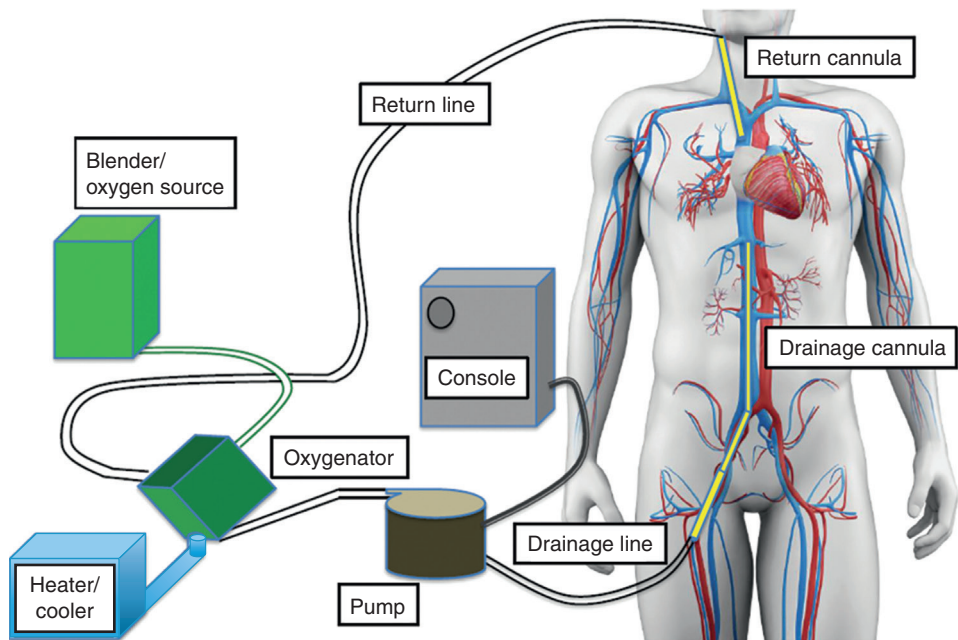


FIG. 9.1 Schematic of circuit components. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

TUBING

The tubing is the conduit by which the blood flows through to the various components, namely the pump and oxygenator. Tubing is made of a flexible, clear, plastic that allows visualization of the color of blood as well as any areas of potential turbulence or clot.

Circuits can be built to different specifications, but the standard for adults usually includes tubing with a diameter of 3/8". This width is wide enough to minimize resistance to flow while not requiring an excessive amount of extracorporeal blood. Some pediatric circuits and low-flow extracorporeal membrane oxygenation (ECMO) circuits may take advantage of smaller tubing such as 1/4", which allows for less circulating blood but may have some impedance of flow.

The length of the tubing can vary – usually the standard is around 15 feet. You may encounter circuits with longer or shorter tubing. The main advantage of longer tubing is less constraint of having to be close to the circuit, but this should be compared to the disadvantages to include more length to monitor, more possibilities for occlusion, and higher priming volume.

3/8" tubing carries about 20 mL per foot, meaning that a circuit with 15 feet of tubing has a priming volume of 300 mL plus the priming volume of the oxygenator (usually around 250–300 mL). The priming volume is important for several reasons:

1. It represents the dilutional volume when going on ECMO.
2. It represents the amount of blood lost when a circuit is lost or changed.
3. It represents the increase in volume of distribution to be considered for pharmacokinetics and effect of medications.

PUMP

We were introduced to the blood pump in [Chapter 6](#), now let's go a little deeper into what it involves. Blood pumps were traditionally roller pumps, which milked the blood forward; however, these have been largely replaced by centrifugal pumps in adults.

Although there are many different models of centrifugal pump, the central mechanism is a cone or a fin that spins around in a centrifugal manner. This pump head is connected to a motor that drives this rotation ([Fig. 9.2](#)).

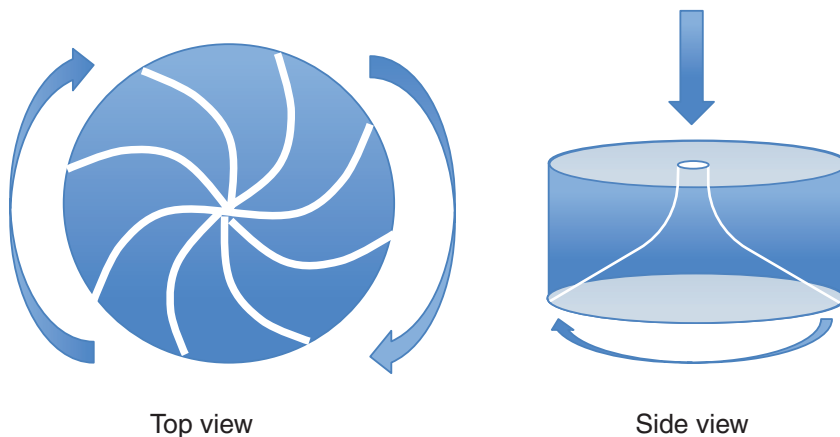


FIG. 9.2 Centrifugal pump

The result is a creation of a vortex of blood that is pulled in through the top and pushed out through the bottom as illustrated in [Fig. 9.3](#).

This pull/push mechanism by which blood enters/exits the pump is an essential part to understanding the flow dynamics related to ECMO, so let's reflect on it.

Because of the rotational mechanism of the centrifugal pump, any blood going **into** the pump will be under **negative** pressure, while any blood **leaving** the pump will be under **positive** pressure.

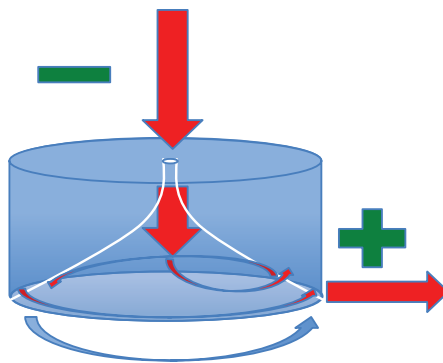


FIG. 9.3 Blood flow through centrifugal pump with negative pressure at the inlet and positive pressure at the outlet

We will go into how this affects the flow of blood in much greater detail in [Chapter 10](#), when we cover flow dynamics. For now, we can start to appreciate how this phenomenon will affect the flow of blood and how we can approach the circuit.

Essence of Centrifugal Pump: Preload Dependence and Afterload Sensitivity

Because the centrifugal pump generates a negative pressure on the inlet and positive pressure on the outlet, you have preload dependence and afterload sensitivity. This means that the availability of blood going to the pump (preload) is requisite for the pump to work, while impedance of flow (afterload) will result in a decrease in the forward flow of blood.

To better visualize afterload sensitivity, think of a boat that is parked in front of a dock. The propeller can spin as fast as you would like; however, if there is a barrier in front of the boat (the dock in this case), there will be no forward movement. The same is true with the centrifugal pump. If you kink or clamp the outflow line, the pump will keep spinning at the same number of revolutions per minute (RPMs), but the flow will drop off.

This represents a major differentiator from roller pumps, where an occlusion downstream from the pump will only cause pressures to increase, as the forward flow of blood will be relatively fixed.

There are two major implications of this afterload sensitivity, the first being that flow will be impacted by the pressure that exists downstream of the pump. This could be anything that impedes flow such as a clot in the oxygenator, a kink in the return line, a bend in the return cannula, or an elevation in the systemic blood pressure if on VA ECMO. The remedy for this would be to fix any of these impediments to flow or to increase the RPMs in an attempt to overcome this afterload and optimize flow ([Fig. 9.4](#)).

The second implication of the afterload sensitivity of the centrifugal pump is that the pump can be subjected to **retrograde flow** at lower RPMs while on VA ECMO. Why is this? Let's consider an example to illustrate.

Say you are decreasing the RPMs down to lower the blood flow on a patient who is slowly recovering with improving blood pressure. As you continue to decrease the RPMs, eventually, the afterload of the patient's mean arterial pressure (MAP) drops the flow to 0. At this point, there exists a higher pressure on the return side of the pump than on the drainage side (as the MAP is higher than the venous pressure), and you will get retrograde flow from the arterial system to the venous system, as illustrated in [Fig. 9.5](#).

The clinical implication of this is that if your RPMs drop below a critical threshold, you can induce a left to right shunt, which can cause a rapid decompensation.

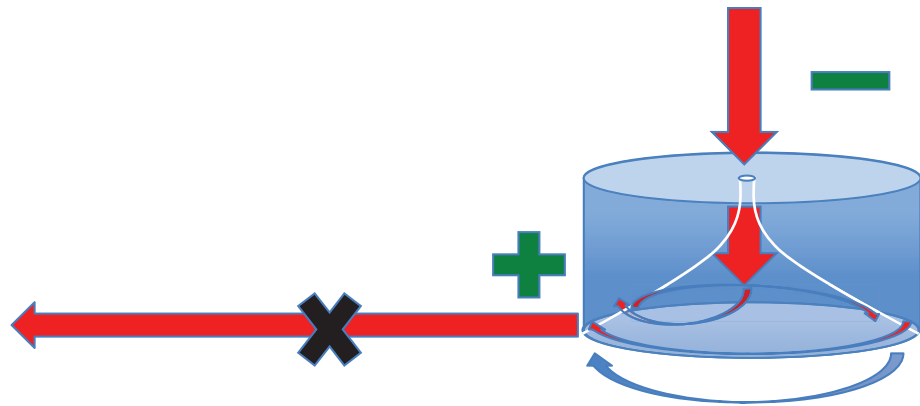


FIG. 9.4 Afterload sensitivity: anything that increases pressure downstream from the pump can impact overall flow

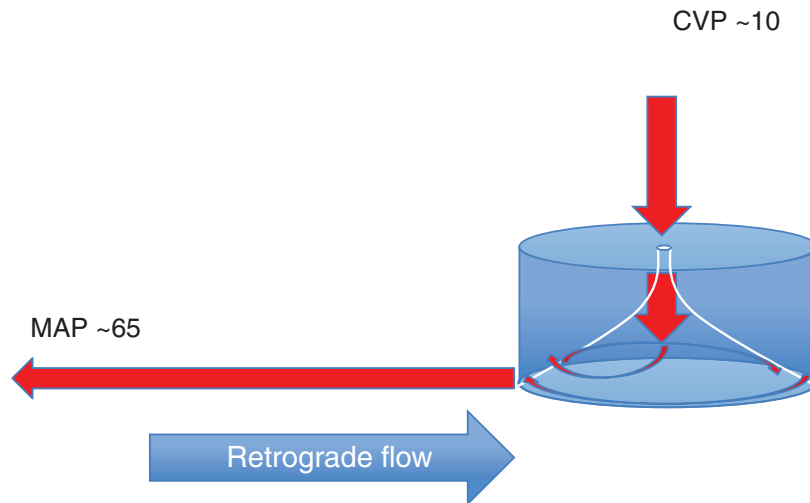


FIG. 9.5 Retrograde flow on VA ECMO with lower pump speed due to differential between arterial and venous pressures. *CVP*, Central venous pressure; *MAP*, mean arterial pressure.

MECHANICAL BACKUP

If our discussion of the pump does anything, it should highlight how bad things can get if the circuit stops flowing. Let's review why:

1. Drop in the flow of oxygenated blood in a patient who may be dependent on the oxygenated blood to maintain delivery of oxygen
2. Lack of blood flow in a patient who may be dependent on that flow to maintain cardiac output
3. Retrograde flow with worsening of the right to left shunt and infusion of venous blood into arterial circulation (on VA ECMO)
4. Stagnation of blood with thrombus formation

Taken together, these may lead to cardiac arrest and rapid deterioration.

Having a strategy to mitigate pump failure is paramount. Ensure there is a backup for power failure, whether it is a hand crank or a backup console next to the pump at all times. This includes

during procedures, transport to anywhere in the hospital, physical therapy, etc. That is one thing about emergencies – they always seem to happen when we are least prepared!

CONSOLE

We discovered during our initial introduction to the console that in its most basic form, it is a mechanism for manipulating RPMs to the pump via connection with the electric motor and a display of flow and RPMs. The mechanism for increasing/decreasing RPMs can be a push button or a dial, with the convention being clockwise rotation to increase RPMs and counterclockwise rotation to decrease RPMs.

You are probably arriving to the conclusion that there are many determinants of blood flow (we will continue to develop these concepts in [Chapter 10](#)). Thus, RPMs are set by the console and transmitted to the motor/pump. The flow that is obtained by these RPMs has to be measured directly.

SENSORS: MEASURING ECMO SUPPORT

The way that blood flow is measured during ECMO support is by direct measurement through sensor probes. This is one advantage of having a support device with blood that is circulated outside the body – by contrast, ventricular support devices determine flow by a calculated algorithm as there is nothing to measure directly.

Flow sensors are circumferential sensors that surround the tubing and are connected to the console. They are placed on the return tubing and must be oriented in the same direction as the blood is flowing. If the sensor probe is on backwards, the console will read a negative flow rate ([Fig. 9.6](#)).

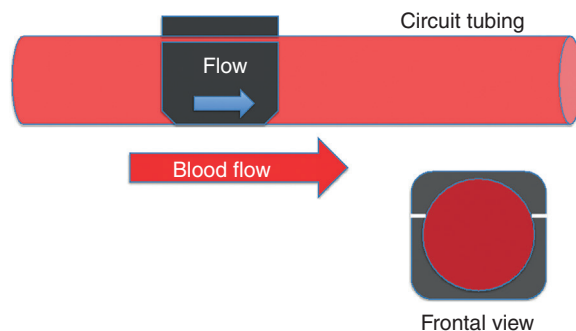


FIG. 9.6 Flow sensor probe

They function by sending sonographic signals to each pole, which are disrupted in proportion to the flow of blood. The faster the flow of blood the greater the reduction of waves moving against that flow and the greater the augmentation of waves moving with the flow of blood as illustrated in [Fig. 9.7](#). The degree of augmentation/reduction of these waves is used to calculate a flow of blood.



FIG. 9.7 Mechanism of flow sensor

Sensors can also be used to measure hemoglobin or the presence of air in the circuit. Air is measured through the disruption of sonographic waves, which normally pass from one side of the sensor to the other uninterrupted (Fig. 9.8).

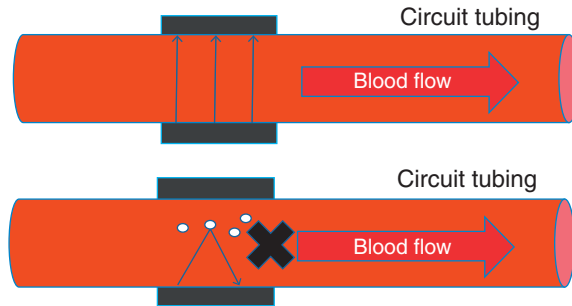


FIG 9.8 Mechanism of air sensor

The effect of this depends on how the console is set, either to alarm or for intervention. As the intervention usually involves stopping the pump, the decision on whether or not interventions should be set involves an evaluation of the risks of air versus the risks of pump stoppage. Regardless, consistency should predominate.

CIRCUIT ACCESS POINTS

At its most basic form, everything we have covered to this point makes for a complete circuit. We have covered all of the essential components. For some programs, this is extent of the circuit. However, most circuits have access points where the circuit can be accessed, which allows for blood draws, medication administration, continuous renal replacement therapy (CRRT) connections, and pressure monitoring. These come in the form of pigtail connections as seen in Fig. 9.9.

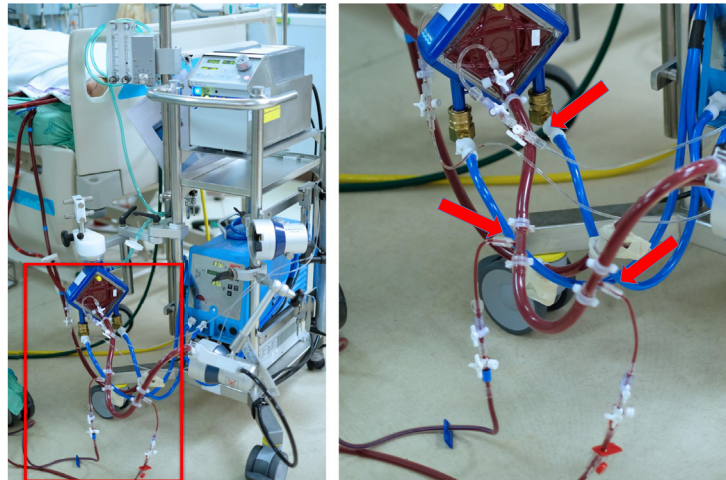


FIG 9.9 Circuit access points. (From PIJITRA PHOMKHAM/Shutterstock.com.)

Let's simplify our circuit illustration, as there really are only a few spots in the circuit where there can be access points: pre-pump, post-pump/pre-oxygenator, and post-oxygenator as illustrated in Fig. 9.10.

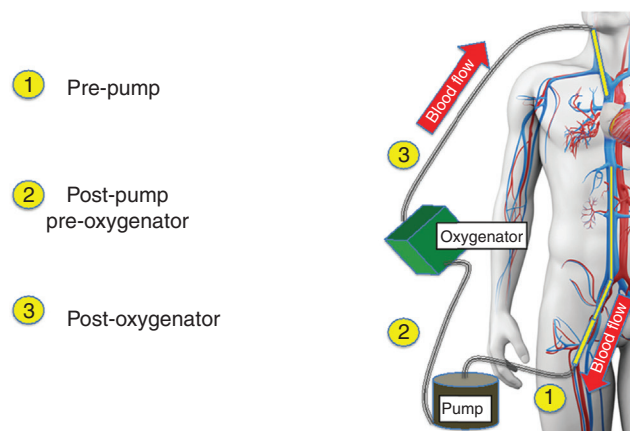


FIG. 9.10 Potential circuit access points. (Modified from SciePro/Shutterstock.com.)

Let's review each individually:

1. **Pre-pump.** Beware of this hookup! Remember, this is under negative pressure, so any time you access it, it air can be entrained. Entrained air can deprime your oxygenator, air lock (and stop) your pump, or embolize to the patient. The negative pressure can allow for quick administration of fluid blood but may be a risky proposition.
2. **Post-pump/pre-oxygenator.** This is your safety zone (or at least your *safer* zone). It is positive pressure, but you have an oxygenator between your access and the patient, which could potentially “catch” any air/clot that gets pushed through the access.
3. **Post-oxygenator.** This access point is also under positive pressure but is a little more risky, since there is no oxygenator between any clot or air that gets administered to the circuit.

Some ECMO programs have multiple points of access to the circuit while some have none. Just remember that the more you access the circuit, the greater the risk for bleeding, alarms, embolization, or potentially air entrainment. Ultimately, it becomes a matter of risk/benefit – what gives enough capabilities but limits the risk of an adverse event.

Let's now go over several indications for accessing the circuit.

Access Point Indication #1: Medication/Blood/Fluid Administration

If you were going to choose one position to administer to the circuit, it would be post-pump/pre-oxygenator (position 2 in Fig. 9.11). This site allows for positive pressure and allows the oxygenator to stand in the way of any air or clot that happens to be entrained. Anticoagulation can also be administered here, which theoretically could help to minimize clot formation on the oxygenator.

Access Point Indication #2: Blood Gas Draws

If you are drawing blood gases for a patient on ECMO, there are three positions from which you will draw: pre-oxygenator, post-oxygenator, and from the patient (blue arrows in Fig. 9.12). We will discuss the rationale more in Chapter 11 as well as some techniques for interpreting these gases. For now, suffice it to say that checking circuit gases both pre-oxygenator and post-oxygenator allows us to diagnose issues and evaluate the circuit.

Access Point Indication #3: Pressure Monitoring

Pressure monitoring can give important data on how to manage the circuit. Pressures can be transduced in any of our positions, and can help to diagnose and manage issues related to the pump. Remember that position 1 is negative pressure, so usually it is only transduced if it is an integrated sensor that minimizes the risk for access to the negative side of the circuit.

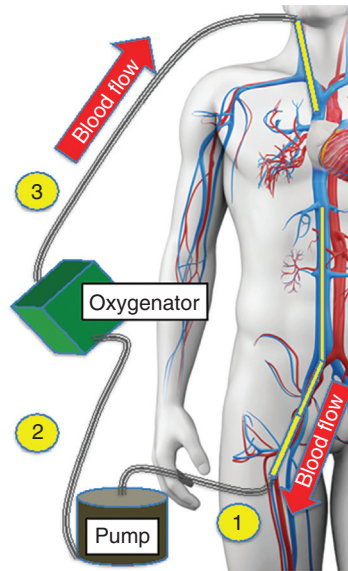


FIG. 9.11 Circuit access for medication/blood/fluid administration. (Modified from [SciePro/Shutterstock.com](#).)

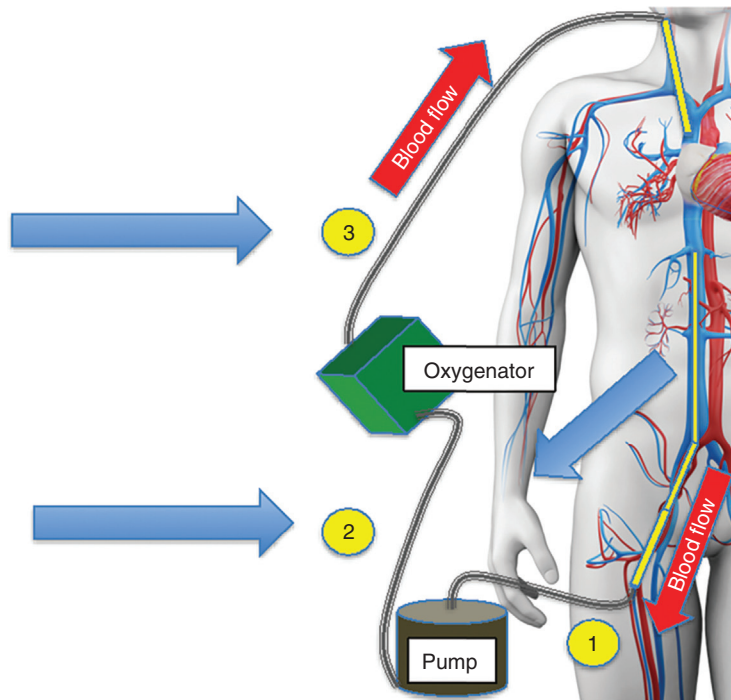


FIG. 9.12 Sites to draw blood gas samples on ECMO: patient arterial line, post oxygenator, and pre oxygenator. (Modified from [SciePro/Shutterstock.com](#).)

The anticipated pressures are as illustrated in [Fig. 9.13](#).

Point 1 is the lowest pressure in the system. Remember that the centrifugal pump generates the flow for the entire system, so you should anticipate the most negative pressure measured to be right before the pump and the highest pressure generated to be right after the pump. Pressure

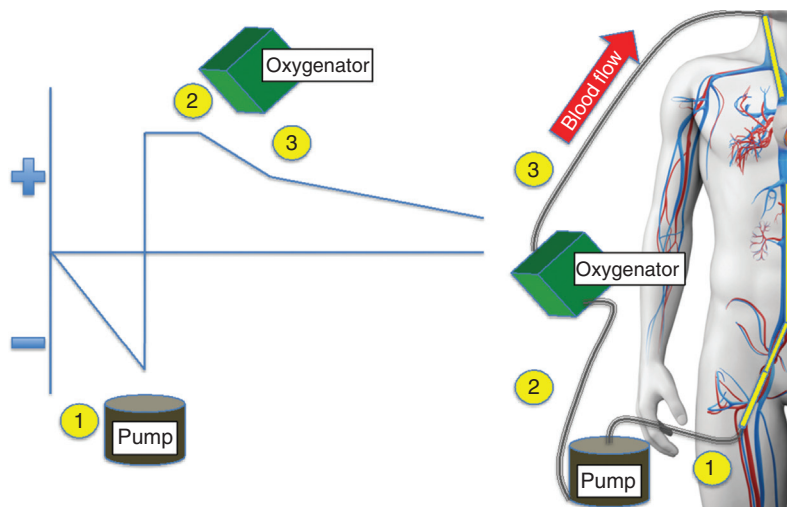


FIG. 9.13 Schematic of circuit pressures relative to ECMO pump and oxygenator. (Modified from [SciePro/Shutterstock.com](#).)

should gradually decrease the further you get from the pump, with a distinct drop-off between point 2 and point 3, which represents the pressure differential across the oxygenator. We will go into the interpretation of these pressure differentials more in the next two chapters.

Access Point Indication #4: CRRT Connection

CRRT and dialysis can be connected directly to the ECMO circuit, obviating the need for a separate catheter. Connecting CRRT can be another point that differs between ECMO programs – some prefer to minimize access to the circuit and always place a separate catheter, some find it a convenient way to initiate renal replacement (Fig. 9.14).

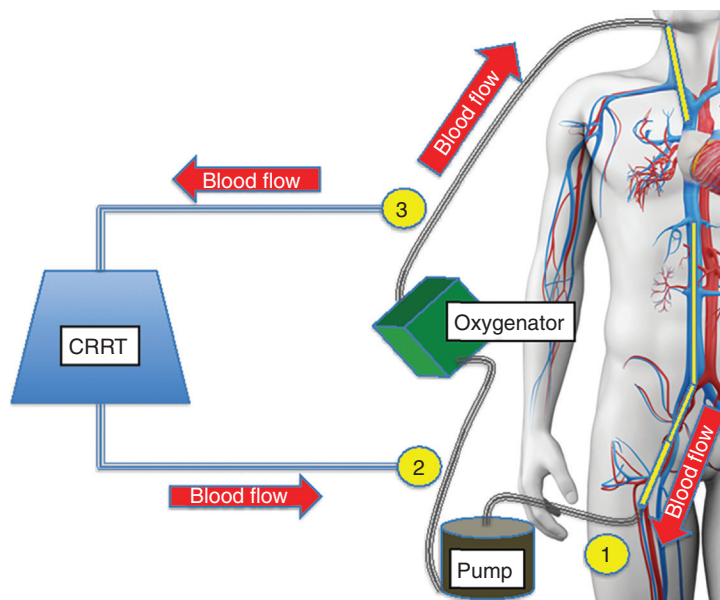


FIG. 9.14 Example connection for continuous renal replacement therapy. (Modified from [SciePro/Shutterstock.com](#).) CRRT, Continuous renal replacement therapy.

Some dialysis machines may work better with different connections based on the pressure differential; however, all else being equal, connecting from post-oxygenator to pre-oxygenator allows an additional element of safety by putting the oxygenator between blood that is returning and the patient, allowing for a potential barrier for any air/thrombus that gets returned by the dialysis circuit.

THE CIRCUIT CHECK: THE MOST IMPORTANT PART OF YOUR DAY

Now that we have gone in depth for all of the components of the ECMO circuit, let's introduce one of the most fundamental practices in the management of any patient on ECMO – the circuit check.

The circuit check involves a tip-to-tip inspection of the entire circuit with attention to all of its components.

A Word of Advice on the Circuit Check

Often, when team members perform circuit checks, they do a quick runthrough of the circuit, making some cursory glances at some components. This approach is sure to miss something. Consider treating the circuit check like a pilot treats her preflight checklist. She may have done it thousands of times but uses a checklist to ensure every part of every component of the list is accounted for. The following is a proposed checklist for how to run a circuit check. Remember, practice doesn't make perfect – perfect practice makes perfect!

Be diligent, be purposeful, be consistent.

Checklist for Circuit Check

- ☐ Cart—at least 6 clamps, full O₂ tank, supplies for emergency circuit change, hand crank/backup console
- ☐ Console—plugged into appropriate electrical receptacle, alarms activated, interventions set as appropriate
- ☐ Oxygen blender—connected to O₂ tank or wall O₂/medical air
- ☐ Heater/cooler—plugged into appropriate electrical receptacle, water level filled, hoses/connectors free of kinks and leaks
- ☐ Drain cannula—dressing clean/dry, suture present and intact, connector intact
- ☐ Drain line—free of clots/fibrin build up/defects
- ☐ Drain line connectors—secure, free of leaks, inspected for fibrin buildup
- ☐ Pump—inflow and outflow free from obstruction/fibrin buildup
- ☐ Pre oxygenator—systematically check for clots/fibrin buildup
- ☐ Pre oxygenator pigtails—patent, stopcocks oriented in correct position, clamps intact
- ☐ Oxygenator—exhaust free from obstruction, green O₂ tubing connected to appropriate gas source, water lines connected and free of leaks
- ☐ Post oxygenator—systematically check for clots and fibrin build up
- ☐ Post oxygenator pigtails—patent, stopcocks oriented in correct position, clamps intact
- ☐ Flow probe/bubble detector—connected, activated as appropriate, free of line defects
- ☐ SvO₂ probe—connected, free of line defects
- ☐ Return line—free of clots/fibrin buildup/defects
- ☐ Return line connectors—secure, free of leaks, inspected for fibrin buildup
- ☐ Return cannula—dressing clean/dry, suture present and intact, connector intact

SUGGESTED READING

- de Tymowski, C., Augustin, P., Houissa, H., Allou, N., Montravers, P., & Delzongle, A., et al. (2017). CRRT connected to ECMO: managing high pressures. *ASAIO Journal*, 63(1), 48–52.
- Lawson, D. S., Ing, R., Cheifetz, I. M., Walczak, R., Craig, D., & Schulman, S., et al. (2005). Hemolytic characteristics of three commercially available centrifugal blood pumps. *Pediatric Critical Care Medicine*, 6(5), 573–577.
- Pedersen, T. H., Videm, V., Svennevig, J. L., Karlsen, H., Ostbakk, R. W., & Jensen, O., et al. (1997). Extracorporeal membrane oxygenation using a centrifugal pump and a servo regulator to prevent negative inlet pressure. *The Annals of Thoracic Surgery*, 63(5), 1333–1339.
- Westrope, C., Harvey, C., Robinson, S., Speggorin, S., Faulkner, G., & Peek, G. J., et al. (2013). Pump controlled retrograde trial off from VA-ECMO. *ASAIO Journal*, 59(5), 517–519.